

## ENG 101

### LESSON 4

#### What is a computer?

- 1- A computer is a machine with an intricate network of electronic circuits that operate switches or magnetize tiny metal cores. The switches, like the cores, are capable of being in one of two possible states, that is, on or off: magnetized or demagnetized. The machine is capable of storing and manipulating numbers, letters, and characters. The basic idea of a computer is that we can make the machine do what we want by inputting signals that turn certain switches on and turn others, off, or that magnetize or do not magnetize the cores.
2. The basic job of computers is the **processing** of information. For this reason, computers can be defined as **devices** which accept information in the form of instructions called a **program** and characters called **data**, perform mathematical and / or logical operations on the information, and then supply results of these operations. The program, or part of it, which tells the computers what to do and the data, which provide the information needed to solve the problem, are kept inside the computer in a place called **memory**.
3. Computers are thought to have many remarkable powers. However, most computers, whether large or small have three basic capabilities. First, Computers have circuits of performing arithmetic operations, such: addition, subtraction, division, multiplication and exponentiation. Second, computer have a means of communicating with the user. After all, if we couldn't feed information in and get results back, these machines wouldn't be of much use. However, certain computers (commonly minicomputers and microcomputers) are used to control directly things such as robots, aircraft's navigation systems, medical instruments, etc.
4. Some of the most common **methods** of **inputting information** are to use **punched cards**, **magnetic tape**, **disks** and **terminals**. The computer's **input device** (which might be a card reader, a tape drive or disk drive, depending on the **medium** used in putting information) reads the information into the computer. For outputting information, two common devices used are a printer which prints the new information on paper, or a **CRT display screen** which shows the results on a TV-like screen.
5. Third, computers have circuits which can make **decisions**. The kinds of decisions which computer circuits can make are not of the type: 'Who would win a war between two countries?' or 'Who is the richest person in the world?' Unfortunately, the computer can only decide three things, namely: Is one number less than another? Are two numbers equal? An, Is one number greater than another?
6. A computer can solve a series of problems and make hundreds, even thousands, of logical decisions without becoming tired or bored. It can find the solution to problem in a fraction of the time it takes a human being to do the job. A computer can replace people in dull, routine tasks, but it has no originality; it works according to the **instructions** given to it and cannot exercise any value judgments. There are times when a computer seems to operate like a mechanical 'brain', but its achievements are limited by the minds of human beings. A computer cannot do anything unless a person tells it what to do and given it the appropriate information; but because electric pulses can move at the speed of light, a computer can carry out vast numbers of arithmetic-logical operations almost instantaneously. A person can do everything a computer can do, but in many cases that person would be dead long before the job was finished.

N. Mullen & P. Brown: *English for Computer Science*; OUP 16-18

#### MAIN IDEA

Now that you have read each paragraph in detail, let us look at the text as a whole. Read the following four statements.

Choose the statement that best expresses the main idea.

1. Computers have changed the way in which many kinds of jobs are done.
2. Instructions and data must be given to the computer to act on.
3. Computers are machines capable of processing and outputting.
4. Without computers, many tasks would take much longer to do.

**UNDERSTANDING THE PASSAGE:**

I shall read out nine statements one by one. Decide if these statements are true or false by referring to the text.

| T                        | F                        |                                                                                                                    |
|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> | 1. A computer can store or handle any data even if it hasn't received information to do so.                        |
| <input type="checkbox"/> | <input type="checkbox"/> | 2. All computers accept and process information in the form of instructions and characters.                        |
| <input type="checkbox"/> | <input type="checkbox"/> | 3. The information necessary for solving problems is found in the memory of the computer.                          |
| <input type="checkbox"/> | <input type="checkbox"/> | 4. Not all computers can perform arithmetic operations, make decisions, communicate in some way with the user.     |
| <input type="checkbox"/> | <input type="checkbox"/> | 5. Computers can still be useful machines even if they can't communicate with the user.                            |
| <input type="checkbox"/> | <input type="checkbox"/> | 6. There are many different devices used for feeding information into a computer.                                  |
| <input type="checkbox"/> | <input type="checkbox"/> | 7. There aren't as many different types of devices used for giving results as there are for accepting information. |
| <input type="checkbox"/> | <input type="checkbox"/> | 8. Computers can make any type of decision they are asked to.                                                      |
| <input type="checkbox"/> | <input type="checkbox"/> | 9. Computers can work endlessly without having to stop to rest unless there is a breakdown.                        |